

Department of Computer Science and Engineering

UPC - Question Bank

Academic Year : 2025 - 2026

Year / Sem : II / III

Subject Code & Name : CS4305 - Python Programming

SLOT	QUESTION	TOTAL MARK	MARK SECURED	SIGN
	UNIT V			
SLOT 1	1. Explain the fundamental principles of GPU architecture and how it enables parallel processing	16		
	2. Explain CUDA programming model with a code snippet	16		
	3. Discuss how to scale arrays with Dask in detail.	12		
	4. Discuss the intersection of AI and parallel computing, highlighting the role of parallelism in accelerating AI algorithms	12		
	5. Describe practical scenarios where parallel computing has been applied to solve complex computational problems.	16		
	6. You are asked to process a 100GB dataset for feature extraction in a machine learning task. Discuss how you would use parallel computing with CUDA or Dask to optimize execution time.	12		
	UNIT IV			
SLOT 2	1. Explain the architecture of a CPU and how it relates to parallel programming. Discuss the significance of cores and threads in this context.	12		
	2. Discuss the Global Interpreter Lock (GIL) in Python. What are its implications for multithreading and how does it impact performance?	12		
	3. Explain the concept of asynchronous programming with AsyncIO. Discuss how AsyncIO enables concurrency and parallelism in Python applications.	12		
	4. Compare and contrast different approaches to achieving parallelism in distributed systems, such as shared-memory vs. message-passing architectures	16		
	5. Describe the organization of memory in a typical computer system, including concepts such as cache, RAM, and virtual memory. Explain how memory organization influences parallel programming	16		
	6. A CPU has 4 cores and supports hyper-threading (2 threads per core). How many threads can it execute simultaneously? Briefly justify your answer.	16		
	UNIT III			
SLOT 3	1. In NumPy, how can Boolean masks be used to replace all negative values in an array with zero?	12		
	2. Explain fancy indexing in NumPy with an example. How does it differ from slicing?	16		
	3. Which Pandas method allows selecting rows by label range and how does it differ from integer position-based selection?	16		

	4. Which SciPy submodule would you use for solving a system of linear equations? Give an example code to solve: $3x+y=9$ $x+2y=8$	12		
	5. Describe how you would implement a bank counter simulation in SimPy where customers arrive at random intervals and there are 2 tellers. Outline the processes and events involved.	16		
	6. Why is it important to split a dataset into training and testing sets before building a model in Scikit-learn?	12		
	UNIT II			
SLOT 4	1. Investigate the roles of the os and sys modules in system-level operations and explain when each should be used.	16		
	2. Write a program that reads text from one file and writes it in binary format to another file	16		
	3. Demonstrate the concept of single inheritance with an example program	12		
	4. Write a program to read data from a CSV file and display it in tabular format.	12		
	5. Analyze the role of Django's MVC architecture in separating concerns and streamlining web development	16		
	6. Demonstrate using the calendar module in Python by writing a script that displays the calendar for a given year	16		
	UNIT I			
SLOT 5	1. Develop a function that takes a string as a parameter and replaces the first letter of every word with the corresponding uppercase letter.	16		
	2. Create a Python program that prompts the user to enter 5 numbers and stores them in a list. Then, print their sum, average, maximum and minimum using built-in functions.	16		
	3. Relate Lists, Tuples, Sets, Frozen Sets, and Dictionaries with examples. Mention at least three key differences for each.	12		
	4. Define mutable and immutable data types with examples.	12		
	5. Discuss the concept of exception handling in Python and how it can be used to handle errors	16		
	6. Describe the following (i) Creating the list (ii) Accessing values in the lists (iii) Updating the list (iv) Deleting the list elements	16		
	2 marks			
	1. Write a code snippet to accept a float and a complex number from the user	2		
	2. Write the significance of indentation in Python code.	2		
	3 What is the role of an assertion statement in Python?	2		
	4 Explain the usage of decorators in Python and provide Examples	2		
	5 How can you create a boolean mask for elements greater than a certain value in a Numpy array?	2		

SLOT 6	6 What is the difference between loc[] and iloc[] in Pandas?	2		
	7 Show how processes communicate using send() and recv() .	2		
	8 What is data parallelism in distributed computing?	2		
	9 What is CUDA and how does it relate to GPU programming?	2		
	10 Why is lazy evaluation important in Dask computations	2		